

Homework 9

All Ungraded

- 1) Consider LAX, a notoriously busy airport with many arriving passengers who want to get to their destinations as soon as possible. There is an available fleet of n Uber drivers to accommodate all passengers who need to be assigned to various intervals of the day. There is a traffic regulation at the airport that limits the total number of Uber drivers at any given hour-long interval to k . Assume that there are p time intervals. Each driver provides a subset of the time intervals he or she can work at the airport, with the minimum requirement of driver j being a_j hour(s) per day and the maximum b_j hour(s) per day. Lastly, the total number of Uber drivers available per day (summed over all intervals) must be at least m to maintain a minimum customer satisfaction and loyalty.
 - a) Design a network flow based algorithm to determine if there is a valid way to schedule the Uber drivers with respect to these constraints.
 - b) Prove correctness of your algorithm.
- 2) Counter Espionage Academy instructors have designed the following problem to see how well trainees can detect SPY's in an $n \times n$ grid of letters S, P, and Y. Trainees are instructed to detect as many disjoint copies of the word SPY as possible in the given grid. To form the word SPY in the grid they can start at any S, move to a neighboring P, then move to a neighboring Y. (They can move north, east, south or west to get to a neighbor.) The following figure shows one such problem on the left, along with two possible optimal solutions with three SPY's each on the right.
 - a) Give an efficient network flow-based algorithm to find the largest number of SPY's.
 - b) Prove correctness of your solution.

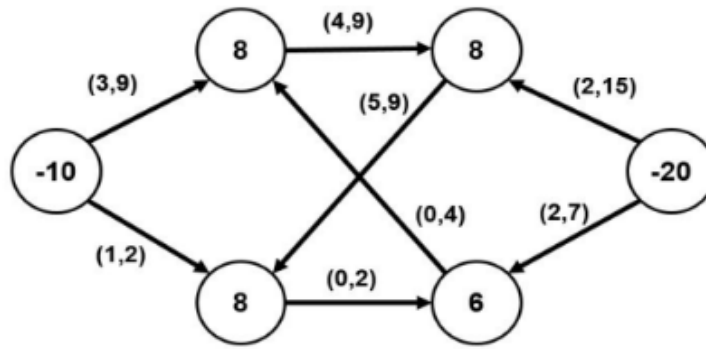
Note: We are only looking for the largest **number** of SPYs, not their actual location.

Y	S	S	P
P	S	P	Y
S	S	Y	P
Y	S	P	S

Y	S	S	P
P	S	P	Y
S	S	Y	P
Y	S	P	S

Y	S	S	P
P	S	P	Y
S	S	Y	P
Y	S	P	S

- 3) In the network below, the demand values are shown on vertices. Lower bounds on flow and edge capacities are shown as (lower bound, capacity) for each edge. Determine if a feasible circulation exists or not. If it does, show the feasible circulation. If it does not, show the cut in the network that is the bottleneck. Show all your steps.



- 4) USC Admissions Center needs your help in planning paths for Campus tours given to prospective students or interested groups. Let USC campus be modeled as a weighted, directed graph G containing locations V connected by one-way roads E . On a busy day, let k be the number of campus tours that have to be done at the same time. It is required that the paths of campus tours do not use the same roads. Let the tour have k starting locations $A = \{a_1, a_2, \dots, a_k\} \subseteq V$. From the starting locations, the groups are taken by a guide on a path through G to some ending location in $B = \{b_1, b_2, \dots, b_k\} \subseteq V$. Your goal is to find a path for each group i from the starting location a_i to any ending location b_j such that no two paths share any edges & no two groups end in the same location b_j .
- (a) Design an algorithm to find k paths $a_i \rightarrow b_j$ that start and end at different vertices, such that they do not share any edges.
- (b) Modify your algorithm to find k paths $a_i \rightarrow b_j$ that start and end in different locations, such that they do not share any edges or vertices.
- 5) Suppose you are on vacation visiting a new city. You will be spending d days there which you want to plan in advance. You have identified k different neighborhoods of the city that you must visit at least once during your trip. Each neighborhood i has R_i different restaurants. Each restaurant offers one of the 5 cuisines you enjoy: Italian, Mexican, Indian, Mediterranean, Chinese. You want to make sure you try at least one restaurant from each cuisine during your vacation, but you don't want to eat at the same exact restaurant more than twice. Your plan for each day consists of visiting a neighborhood and picking one restaurant for dinner within that neighborhood.
- a) Devise a network flow based efficient algorithm to determine if such a vacation plan is feasible.
- b) Prove the correctness of your solution.

6) There is a precious diamond on display in a museum. After a recent museum robbery, it was decided to tighten the security arrangements. There are n security guards available who can be deployed for the task. The diamond is to be guarded at m disjoint time intervals of the day, and must have either one or two guards deployed in each interval. Each guard provides a list of intervals for which they are available to be deployed. Work regulations require that the deployment for each guard must be for at most M and at least L time slots.

- a) Design a network flow algorithm to determine a feasible assignment of the guards given the constraints above, and output “no feasible assignment” if it does not exist.
- b) Prove correctness of the algorithm.